# KARTHIKEYA NAGISETTI

AI Engineer · Machine Learning Systems · LLM Fine-Tuning & RAG

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## SUMMARY

AI Engineer at Meta shipping production LLM systems end-to-end — agent runtimes, hybrid RAG, eval harnesses, MCP tool servers — **200+ landed changes across ~190K lines in 9 months**. 3+ years from PyTorch training loops to Triton inference and on-call: fine-tuning on 8× A100s, fraud detection at ~600 req/sec at p99 < 120 ms. **Author of Snap-Time Agent** — an open-source Bounded-Autonomy framework for LLM agents.

## TECHNICAL SKILLS

**ML / DL:** PyTorch, Hugging Face Transformers, TRL, LoRA / QLoRA, DeepSpeed (ZeRO-2/3), ONNX, scikit-learn, XGBoost
**LLMs & RAG:** LLaMA 2/3, Mistral, GPT-4 / Claude APIs, prompt engineering, vector retrieval (pgvector, FAISS), eval harnesses, guardrails, MCP tool servers, LangGraph
**Languages:** Python (expert), SQL, Bash, C++ (intermediate), TypeScript
**Infrastructure:** Docker, Kubernetes, AWS (S3, EC2, SageMaker), GCP (Vertex), NVIDIA Triton Inference Server, Linux
**MLOps:** MLflow, Weights & Biases, Optuna, Airflow, GitHub Actions, Terraform (basic), shadow A/B testing
**Data:** PostgreSQL (+ pgvector), Redis, Spark / PySpark, DuckDB, Pandas, NetworkX, DGL

## EXPERIENCE

**Meta** — AI Engineer — Menlo Park, CA · Dec 2025 – Present

- Shipped **200+ landed code changes across ~190K lines in 9 months as a single IC** on **Deployment Buddy**, Meta's internal LLM-powered assistant on LLaMA 3.x — across backend services, LLM pipelines, data infrastructure, dashboards, and native mobile.
- Built and scaled an **AI-powered deployment automation platform** (19 operational workflows, 650K+ proactive notifications, 125K+ resulting updates) that saved **6,600+ manual hours and delivered ~$2.9M in operational value within six months**.
- Architected the **agent runtime** — lazy tool loading, sub-agent SQL verification, message-list compaction, cross-session memory — **cutting context-token usage up to 71%** while raising answer reliability on long multi-turn sessions.
- Implemented a **hybrid retrieval pipeline** (vector + BM25) over internal docs; **cut average context-lookup time ~40%** vs. the previous prototype. Built a **regression-eval harness** on a held-out 1.2K-prompt set that surfaced two recurring hallucination patterns driving prompt-and-retrieval changes.
- Built an **LLM classification pipeline** coding operational free-text into a governed taxonomy at **99%+ coverage and 0.88 average confidence**, with feature-flagged prompt versions and an A/B parity rig; and an **incident-linking system** at **~96% human acceptance** across thousands of evaluations.
- Shipped **write-capable automation** with enqueue-time authorization, pre-write revalidation, idempotency keys, and dual kill switches; exposed the query and mutation layer as **MCP tools** across 5+ AI clients, taking a cold multi-second query to **~5 ms warm**. Active in code review (Diffs, Sapling) and on-call for the inference service.

**Healed Hearts** — Lead AI Engineer — San Francisco, CA · Jun 2025 – Nov 2025

- Fine-tuned **Llama-2-13B and Mistral-7B** on a 200K-sample clinical-notes dataset using LoRA + 4-bit QLoRA on 8× A100s; reached **87% accuracy on a USMLE-Step-1-style multiple-choice eval** (vs. 81% base).
- Built a **clinical RAG pipeline** (Postgres + pgvector over PubMed abstracts) behind a FastAPI service serving **~5K daily requests at p95 ~280 ms**.
- **Caught hallucinated citations before release** with a Constitutional-AI-style critic loop and an eval framework; ran initial DP-SGD experiments for a privacy-preserving partner-data fine-tune.

**U.S. Bank** — AI Engineer — San Francisco Bay Area · Jun 2024 – May 2025

- Shipped a **fraud-detection model** combining XGBoost with graph features (NetworkX → DGL); **cut false positives ~22%** at the precision target (99.4%) on the production stream.
- Migrated a legacy NLP pipeline (regex + scikit-learn) to a fine-tuned **DistilBERT classifier** for financial-document categorization; **+9 F1 points** and ~3× faster inference after ONNX export.
- Stood up model serving on Kubernetes (Triton + FastAPI gateway) at **~600 req/sec, p99 < 120 ms**; wrote MLflow + Airflow pipelines for nightly retraining and shadow A/B, **cutting the release cycle from ~3 weeks to ~3 days**.

**Ernst & Young LLP** — AI Research Associate — Bengaluru, India · Feb 2023 – Jul 2023

- Built NLP pipelines (FinBERT, RoBERTa) for financial-filing analysis — **~91% F1** on document-section classification; contributed to a fraud-detection prototype (XGBoost + graph features), **+12% recall** over baseline; built an internal Optuna wrapper for hyperparameter sweeps **adopted by 4 team projects**.

## INDEPENDENT RESEARCH & PROJECTS

- **Answer-First Question Discovery** (independent research, in progress) — Inverts the standard QA setup: given an answer, an LLM agent searches for questions it resolves that are *not yet known* to have it, surfacing undiscovered question–answer pairs through an autonomous daily research loop.
- **Snap-Time Agent — Human-Like AI** (independent research) — Open-source **Bounded-Autonomy framework**: LLM agents with multidimensional emotion vectors, importance-weighted temporal memory, and a Snap-Time decision protocol for behavior under deadline pressure. **github.com/25karthikeyanagisetti-code/snap-time-agent**

## EDUCATION

**M.S., Technology Management**, Lindsey Wilson College · GPA 3.8 — Columbia, KY · 2023 – 2025

**B.S., Electrical Engineering**, Vardhaman College of Engineering · GPA 3.2 — Hyderabad, India · 2019 – 2023

**Certifications:** AWS Machine Learning – Specialty · Google Professional ML Engineer